

Custom Messages/Services

Define and build your messages/services as usual. In order Pycharm to recognize them (for autocompletion, etc.):

- Open File – Settings
- Select Project: projectname – Project Structure
- Select src folder and click on Mark as Sources
- Click OK

IDES FOR C++

1. Eclipse

Eclipse's built-in C++ indexing capabilities have gotten quite good in recent versions.

Installing Eclipse

To use this tutorial, users should not “sudo apt-get install eclipse”. Instead:

- Go to eclipse web site
- Click on “download now” from the top-right corner
- Download eclipse for C/C++ developers
- Extract eclipse into a folder of your choice

To make a nice launcher for Ubuntu, google for instructions or check e.g. <http://www.blogs.digitalworlds.net/softwarenotes/?p=54>

Creating the Eclipse project files

1. For a rosbuilt package

CMake can produce Eclipse project files automatically. These project files then set up all include paths correctly, so that auto-completion and code browsing will work out of the box.

However, currently, a small hack is required for generating these project files in the right folder. The problem is that ROS creates the Makefiles and all other build artifacts in the build/ folder, but Eclipse expects the source files to be located within (or below) the folder where its project files are stored.

Fortunately, there is now a make target using a small script that circumvents this problem, by forcing CMake to create the project files in the directory of your package (and not the build/ folder). Proceed as follows:

- Open a terminal, roscd into the folder of your package, and execute: make eclipse-project